Exercise 15D

Solution 1:

(i) Radius of the sphere $= 3.5$ cm

$$\text{Now, volume} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 3.5$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{35}{10} \times \frac{35}{10} \times \frac{35}{10}$$

$$= \frac{539000}{3 \times 1000}$$

$$= \frac{539}{3}$$

$$= 179.67 \text{ cm}^3$$

$\therefore$ Surface area $= 4\pi r^2$

$$= 4 \times \frac{22}{7} \times 3.5 \times 3.5$$

$$= 4 \times \frac{22}{7} \times \frac{35}{10} \times \frac{35}{10}$$

$$= 4 \times 22 \times \frac{5}{10} \times \frac{35}{10}$$

$$= 4 \times 22 \times \frac{1}{2} \times \frac{7}{2}$$

$$= 22 \times 7$$

$$= 154 \text{ cm}^2$$

(ii) Radius of the sphere $= 4.2$ cm

$$\text{Now, volume} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 4.2 \times 4.2 \times 4.2$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{42}{10} \times \frac{42}{10} \times \frac{42}{10}$$

$$= 4 \times 22 \times \frac{14}{10} \times \frac{6}{10} \times \frac{42}{10}$$

$$= 4 \times 22 \times \frac{14}{10} \times \frac{2}{10} \times \frac{42}{10}$$

$$= \frac{310464}{1000}$$

$$= 310.464 \text{ cm}^3$$

$$= 310.46 \text{ cm}^3$$

$\therefore$ Surface area $= 4\pi r^2$

$$= 4 \times \frac{22}{7} \times 4.2 \times 4.2$$

$$= 4 \times \frac{22}{7} \times \frac{42}{10} \times \frac{42}{10}$$

$$= 4 \times 22 \times \frac{6}{10} \times \frac{42}{10}$$

$$= \frac{22176}{100}$$

$$= 221.76 \text{ cm}^2$$

(iii) Radius of sphere=5 m

Now, volume $= \frac{4}{3}\pi r^3$

$$= \frac{4}{3} \times \frac{22}{7} \times 5^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 125$$

$$= \frac{11000}{21}$$

$$= 523.81 \text{ cm}^3$$

$\therefore$ Surface area $= 4\pi r^2$

$$= 4 \times \frac{22}{7} \times 5^2$$

$$= 314.29 \text{ cm}^2$$

Solution 2: Volume of the sphere $= 38808\ \text{cm}^3$

Suppose that $r$cm is the radius of the given sphere.

$$\therefore \frac{4}{3}\pi r^3 = 38808$$

$$\Rightarrow r^3 = \frac{38808 \times 3 \times 7}{4 \times 22}$$

$$\Rightarrow r^3 = \frac{1764 \times 3 \times 7}{4}$$

$$\Rightarrow r^3 = 441 \times 3 \times 7$$

$$\Rightarrow r^3 = 9261$$

$$\Rightarrow r = 21\ cm$$

$\therefore$ Surface area of the sphere $= 4\pi r^2$

$$= 4 \times \frac{22}{7} \times 21 \times 21$$

$$= 4 \times 22 \times 3 \times 21$$

$$= 5544\ \text{cm}^2$$

Solution 3: Volume of the sphere $= 606.375\ \text{m}^3$

Then

$$\frac{4}{3}\pi r^3 = 606.375$$

$$\Rightarrow r^3 = \frac{606.375 \times 3 \times 7}{4 \times 22}$$

$$\Rightarrow r^3 = 144.703$$

$$\Rightarrow r = 5.25\ m$$

$\therefore$ Surface area $= 4\pi r^2$

$$= 4 \times \frac{22}{7} \times 5.25 \times 5.25$$

$$= 346.5\ \text{m}^2$$

Solution 4: Let the radius of the sphere be $r$cm.

Surface area of the sphere $= 154\ \text{cm}^2$

$\therefore 4\pi r^2 = 154$

$\Rightarrow 4 \times \frac{22}{7} \times r^2 = 154$

$\Rightarrow r^2 = \frac{154 \times 7}{4 \times 22}$

$\Rightarrow r^2 = \frac{49}{4}$

$\Rightarrow r = \frac{7}{2}\ \text{cm}^2$

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2}$

$= \frac{1}{3} \times 11 \times 7 \times 7$

$= \frac{539}{3}$

$= 179.67\ \text{cm}^3$

Thus, the volume of the sphere is approximately 179.67 cm^3.

Solution 5: Surface area of the sphere $= (576\pi)\ \text{cm}^2$

Suppose that r cm is the radius of the sphere.

Then

$4\pi r^2 = 576\pi$

$\Rightarrow r^2 = \frac{576}{4}$

$\Rightarrow r^2 = 144$

$\Rightarrow r = 12\ cm$

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \pi \times 12 \times 12 \times 12$

$= 4 \times \pi \times 4 \times 12 \times 12$

$= 2304\pi\ \text{cm}^3$

Solution 6: Here, $l = 12$ cm, $b = 11$ cm and $h = 9$ cm

Volume of the cuboid $= l \times b \times h = 12 \times 11 \times 9$

Radius of one lead shot $= \frac{3}{2}$ mm $= \frac{0.3}{2 \times 10}$ cm $= \frac{0.3}{2}$ cm

Volume of one lead shot $= \frac{4}{3} \times \frac{22}{7} \times \left(\frac{0.3}{2}\right)^3$

$\therefore$ Number of lead shots $= \frac{\text{volume of the cuboid}}{\text{volume of one lead shot}}$

$= \frac{12 \times 11 \times 9}{\frac{4}{3} \times \frac{22}{7} \times \left(\frac{0.3}{2}\right)^3}$

$= \frac{12 \times 11 \times 9}{\frac{4}{3} \times \frac{22}{7} \times \frac{0.027}{8}}$

$= \frac{12 \times 11 \times 9 \times 3 \times 7 \times 8}{4 \times 22 \times 0.027}$

$= 84000$

Solution 7: Radius of the sphere $= 8$ cm

Volume of the sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times 8^3$

$= 2145.52$ cm^3

Radius of one lead ball $= 1$ cm

Volume of one lead ball $= \frac{4}{3} \times \frac{22}{7} \times 1^3 = 4.19$ cm^3

$\therefore$ Number of lead balls $= \frac{\text{volume of the sphere}}{\text{volume of one lead ball}}$

$= \frac{2145.52}{4.19}$

$= 512.05$

≈ 512

Solution 8: Radius of the solid sphere $= 3$ cm

Volume of the solid sphere $= \frac{4}{3}\pi r^3$

$$= \frac{4}{3}\pi \times 3^3 \text{ cm}^3$$

$$= \frac{4}{3}\pi \times 27 \text{ cm}^3$$

Diameter of the spherical ball $= 0.6$ cm

$\therefore$ Radius of the spherical ball $= 0.3$ cm

Volume of the spherical ball $= \frac{4}{3}\pi \times (0.3)^3 \text{ cm}^3$

Now, number of small spherical balls $= \frac{\text{volume of the sphere}}{\text{volume of the spherical ball}}$

$$= \frac{\frac{4}{3}\pi \times 27}{\frac{4}{3}\pi \times (0.3)^3}$$

$$= 1000$$

$\therefore$ The number of small balls thus obtained is 1000

Solution 9: Radius of the metallic sphere, $r_1 = 10.5$ cm

Volume of the sphere $= \frac{4}{3}\pi r^3$

Radius of each smaller cone, $r_2 = 3.05$ cm

Height of each smaller cone $= 3$ cm

Volume of each smaller cone $= \frac{1}{3}\pi r_2^2 h$

Number of cones obtained $= \frac{\text{volume of the sphere}}{\text{volume of each smaller cone}}$

$$= \frac{\frac{4}{3}\pi r_1^3}{\frac{1}{3}\pi r_2^2 h}$$

$$= \frac{4r_1^3}{r_2^2 h}$$

$$= \frac{4 \times 10.5 \times 10.5 \times 10.5}{3.5 \times 3.5 \times 3}$$

$=126.006$

≈ 126

$\therefore$ 126 cones are obtained.

Solution 10: Diameter of each sphere, $d = 12$ cm
Radius of each sphere, $r = 12/2 = 6$ cm

$\therefore$ Volume of each sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(6)^3$ cm^3

Diameter of base of the cylinder, $D = 8$ cm
Radius of base of cylinder, $R = 4$ cm
Height of the cylinder, $h = 90$ cm

$\therefore$ Volume of the cylinder $= \pi R^2 h = \pi(4)^2 \times 90$ cm^3

$$\text{Number of spheres} = \frac{\text{volume of the cylinder}}{\text{volume of the sphere}}$$

$$= \frac{\pi R^2 h}{\frac{4}{3}\pi r^3}$$

$$= \frac{R^2 h}{\frac{4}{3} r^3}$$

$$= \frac{4^2 \times 90 \times 3}{4 \times 6^3}$$

$$= \frac{12 \times 90}{216}$$

$= 5$

$\therefore$ Five spheres can be made.

Solution 11: Let the length of the wire be h cm.
Radius of the wire, $r = 1$ mm $= 0.1$ cm
Radius of the sphere, $R = 3$ cm
Now, volume of the sphere = volume of the cylindrical wire

$$\Rightarrow \frac{4}{3}\pi R^3 = \pi r^2 h$$

$\Rightarrow \frac{4}{3}R^3 = r^2h$

$\Rightarrow 4 \times 3^2 = (0.1)^2 \times h$

$\Rightarrow h = \frac{4 \times 9}{0.1 \times 0.1}$

$\Rightarrow h = 3600\ cm$

$\Rightarrow h = 36\ m$

$\therefore$ Length of the wire $= 36$ m

Solution 12: Radius of the copper sphere, $R = 9$ cm
Length of the wire, $h = 108$ m $= 10800$ cm
Suppose that r cm is the radius of the wire.

Volume of the sphere = volume of the wire

$\frac{4}{3}\pi R^3 = \pi r^2 h$

$\Rightarrow \frac{4}{3}R^3 = r^2h$

$\Rightarrow \frac{4}{3} \times 9^3 = r^2 \times 10800$

$\Rightarrow r^2 = \frac{4 \times 729}{3 \times 10800}$

$\Rightarrow r^2 = \frac{4 \times 81}{3 \times 1200}$

$\Rightarrow r^2 = \frac{9}{100}$

$\Rightarrow r = \sqrt{\frac{9}{100}}$

$\Rightarrow r = \frac{3}{10}$

$\Rightarrow r = 0.3\ cm$

$\therefore$ Diameter of the wire $= 0.6$ cm

Solution 13: Radius of the sphere, r = 7.8 cm
Height of the cone, h = 31.2 cm
Suppose that R cm is the radius of the cone.

Now, volume of the sphere = volume of the cone

$$\Rightarrow \frac{4}{3}\pi r^3 = \frac{1}{3}\pi R^2 h$$

$$\Rightarrow 4r^3 = R^2 h$$

$$\Rightarrow 4 \times (7.8)^3 = R^2 \times 31.2$$

$$\Rightarrow R^2 = \frac{4 \times 7.8 \times 7.8 \times 7.8}{31.2}$$

$$\Rightarrow R^2 = 60.84$$

$$\Rightarrow R = 7.8\ cm$$

∴ The diameter of the cone is 15.6 cm.

Solution 14: Radius of the spherical cannonball, $R = 14$ cm
Radius of the base of the cone, $r = 17.5$ cm
Let h cm be the height of the cone.

Now, volume of the sphere = volume of the cone

$$\Rightarrow \frac{4}{3}\pi R^3 = \frac{1}{3}\pi r^2 h$$

$$\Rightarrow 4R^3 = r^2 h$$

$$\Rightarrow 4 \times (14)^3 = (17.5)^2 \times h$$

$$\Rightarrow h = \frac{4 \times 14 \times 14 \times 14}{17.5 \times 17.5}$$

$$\Rightarrow h = 35.84\ cm$$

∴ The height of the cone is 35.84 cm.

Solution 15: Radius of the original spherical ball, R = 3 cm
The radii of two of these balls are 1.5 cm and 2 cm
Suppose that the radius of third ball is r cm.

Then volume of the original spherical ball = volume of the three spherical balls

$$\Rightarrow \frac{4}{3}\pi R^3 = \left(\frac{4}{3}\pi \times 1.5^3\right) + \left(\frac{4}{3}\pi \times 2^3\right) + \left(\frac{4}{3}\pi \times r^3\right)$$

$$\Rightarrow \frac{4}{3}\pi R^3 = \frac{4}{3}\pi[(1.5)^3 + 2^3 + r^3]$$

$$\Rightarrow R^3 = (1.5)^3 + 2^3 + r^3$$

$\Rightarrow 3^3 = 3.375 + 8 + r^3$

$\Rightarrow 27 = 11.375 + r^3$

$\Rightarrow r^3 = 27 - 11.375$

$\Rightarrow r^3 = 15.625$

$\Rightarrow r = 2.5\ cm$

$\therefore$ The radius of the third ball is 2.5 cm.

Solution 16: Suppose that the radii are r and 2r.

Now, ratio of the surface areas $= \frac{4\pi r^2}{4\pi(2r)^2} = \frac{r^2}{(2r)^2} = \frac{r^2}{4r^2} = \frac{1}{4} = 1:4$

$\therefore$ The ratio of their surface areas is 1 : 4.

Solution 17: Suppose that the radii of the spheres are r and R.

We have:

$\frac{4\pi r^2}{4\pi R^2} = \frac{1}{4}$

$\Rightarrow \frac{r^2}{R^2} = \frac{1}{4}$

$\Rightarrow \frac{r}{R} = \sqrt{\frac{1}{4}}$

$\Rightarrow \frac{r}{R} = \frac{1}{2}$

Now, ratio of the volumes $= \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = \left(\frac{r}{R}\right)^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$\therefore$ The ratio of the volumes of the spheres is 1 : 8.

Solution 18: Suppose that the radius of the ball is r cm.

Radius of the cylindrical tub, R = 12 cm

Water raised in the tub, h = 6.75 cm

Now, volume of the ball = volume of water raised in the cylinder

$\Rightarrow \frac{4}{3}\pi r^3 = \pi R^2 h$

$$\Rightarrow \frac{4}{3}r^3 = (12)^2 \times 6.75$$

$$\Rightarrow \frac{4}{3}r^3 = 144 \times 6.75$$

$$\Rightarrow r^3 = \frac{144 \times 6.75 \times 3}{4}$$

$$\Rightarrow r^3 = 36 \times 6.75 \times 3$$

$$\Rightarrow r^3 = 729$$

$$\Rightarrow r = 9\ cm$$

$\therefore$ The radius of the ball is 9 cm.

Solution 19: Let h cm be the increase in the level of water.

Radius of the cylindrical bucket, r = 15 cm

Height up to which water is being filled = 20 cm

Radius of the spherical ball, R = 9 cm

Now, volume of the sphere = increased in volume of the cylinder

$$\Rightarrow \frac{4}{3}\pi R^3 = \pi r^2 h$$

$$\Rightarrow \frac{4}{3}R^3 = r^2 h$$

$$\Rightarrow \frac{4}{3} \times 9^3 = (15)^2 \times h$$

$$\Rightarrow \frac{4}{3} \times 729 = 225 \times h$$

$$\Rightarrow h = \frac{4 \times 729}{3 \times 225}$$

$$\Rightarrow h = \frac{4 \times 27}{25}$$

$$\Rightarrow h = 4.32\ cm$$

$\therefore$ The increase in the level of water is 4.32 cm.

Solution 20: Outer radius of the spherical shell = 6 cm

Inner radius of the spherical shell = 4 cm

Volume of metal contained in the shell $= \frac{4}{3} \times \frac{22}{7}(6^3 - 4^3)$

$= \frac{88}{21} \times (216 - 64)$

$= \frac{88}{21} \times 152$

$= 636.95 \text{ cm}^3$

$\therefore$ Outer surface area $= 4 \times \frac{22}{7} \times 6 \times 6$

$= \frac{3168}{7}$

$= 452.57 \text{ cm}^2$

Solution 21: Internal radius of the hollow spherical shell, $r = 8$ cm
External radius of the hollow spherical shell, $R = 9$ cm

Volume of the shell $= \frac{4}{3}\pi(R^3 - r^3)$

$= \frac{4}{3}\pi(9^3 - 8^3)$

$= \frac{4}{3} \times \frac{22}{7} \times (729 - 512)$

$= \frac{4 \times 22 \times 217}{21}$

$= \frac{88 \times 31}{3}$

$= \frac{2728}{3} \text{ cm}^3$

Weight of the shell = volume of the shell $\times$ density per cubic cm

$= \frac{2728}{3} \times 4.5$

$= 4092 \text{ g}$

$= \frac{4092}{1000} \text{ g}$

$= 4.092 \text{ kg}$

$\therefore$ Weight of the shell $= 4.092$ kg

Solution 22: Radius of the hemisphere, R = 9 cm

Height of the right circular cone, h = 72 cm

Suppose that the radius of the base of the cone is r cm.

Volume of the hemisphere = volume of the cone

$$\Rightarrow \frac{2}{3}\pi R^3 = \frac{1}{3}\pi r^2 h$$

$$\Rightarrow \frac{2}{3}\pi \times 9^3 = \frac{1}{3}\pi \times r^2 \times 72$$

$$\Rightarrow r^2 = \frac{2 \times 9 \times 9 \times 9}{72}$$

$$\Rightarrow r^2 = \frac{81}{4}$$

$$\Rightarrow r = \frac{9}{2}$$

$$\Rightarrow r = 4.5\ cm$$

∴ The radius of the base of the cone is 4.5 cm.

Solution 23: Internal radius of the hemispherical bowl, R = 9 cm

Radius of a cylindrical shaped bottle, r = 1.5 cm

Height of a bottle, h = 4 cm

$$\text{Number of bottles required to empty the bowl} = \frac{\text{volume of the hemispherical bowl}}{\text{volume of a cylindrical shaped bottle}}$$

$$= \frac{\frac{2}{3}\pi R^3}{\pi r^2 h}$$

$$= \frac{\frac{2}{3}R^3}{r^2 h}$$

$$= \frac{\frac{2}{3} \times 9 \times 9 \times 9}{1.5 \times 1.5 \times 4}$$

$$= \frac{2 \times 3 \times 9 \times 9}{1.5 \times 1.5 \times 4}$$

$$= 54$$

∴ 54 bottles are required to empty the bowl.

Solution 24: Inner radius of the hemispherical bowl, r = 4 cm

Thickness of the bowl $= 0.5$ cm

Now, outer radius of the bowl, R $= (4 + 0.5)$ cm $= 4.5$ cm

Now,

volume of steel used = volume of the outer hemisphere − volume of the inner hemisphere

$$= \frac{2}{3}\pi R^3 - \frac{2}{3}\pi r^3$$

$$= \frac{2}{3}\pi(R^3 - r^3)$$

$$= \frac{2}{3}\pi[(4.5)^3 - (4)^3]$$

$$= \frac{2}{3}\pi(91.125 - 64)$$

$$= \frac{2}{3} \times \frac{22}{7} \times 27.215$$

$$= \frac{44}{21} \times 27.125$$

$$= 56.83 \text{ cm}^3$$

$\therefore$ 56.83 cm^3 of steel is used in making the bowl.

Solution 25: Inner radius of the bowl, $r = 5$ cm

Let the outer radius of the bowl be R cm.

Thickness of the bowl $= 0.25$ cm (Given)

$\therefore$ Outer Radius $= R = 0.25 + r$

$$= 0.25 + 5$$

$$= 5.25 \text{ cm}$$

$\therefore$ Outer curved surface area of the bowl $= 2\pi r^2$

$$= 2 \times \frac{22}{7} \times (5.25)^2$$

$$= 173.25 \text{ cm}^2$$

Thus, the outer curved surface area of the bowl is 173.25 cm^2.

Solution 26: Inner diameter of the hemispherical bowl $= 10.5$ cm

$\therefore$ Inner radius of the bowl, $r = \frac{10.5}{2} = 5.25$ cm

Now,

Inner curved surface area of the bowl $= 2\pi R^2$

$$= 2 \times \frac{22}{7} \times (5.25)^2$$
$$= 173.25 \text{ cm}^2$$

Rate of tin-plating = ₹ 32 per 100 cm^2

∴ Cost of tin-plating the bowl on the inside

= Inner curved surface area of the bowl × Rate of tin-plating

$$= 173.25 \times \frac{32}{100}$$

= ₹ 55.44

Thus, the cost of tin-plating the bowl on the inside is ₹ 55.44.

Solution 27: Let the radius of the moon and earth be r units and R units, respectively.

Now, according to the question

$$2r = \frac{1}{4} \times 2R$$

$$\Rightarrow r = \frac{R}{4} \quad \ldots\ldots (i)$$

$$\therefore \frac{\text{Volume of the moon}}{\text{Volume of the earth}} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3}$$

$$= \frac{r^3}{R^3}$$

$$= \frac{\left(\frac{R}{4}\right)^3}{R^3}$$

$$= \frac{\frac{R^3}{64}}{R^3}$$

$$= \frac{1}{64}$$

Thus, the volume of the moon is $\frac{1}{64}$ of the volume of the earth.

Solution 28: Let the radius of the solid hemisphere be r units.

Numerical value of surface area of the solid hemisphere = $3\pi r^2$

It is given that the volume and surface area of the solid hemisphere are numerically equal.

$\therefore \frac{2}{3}\pi r^3 = 3\pi r^2$

$\Rightarrow 2r = 9$ units

Thus, the diameter of the hemisphere is 9 units.